

Question Number	Answer	Additional guidance	Mark
8(a)(i)	An answer that includes one of the following points: • (errors in genetic sequence of gene) occur in crossing over (1) • the chiasmata may be at slightly different points (in the gene) (1)	accept formation of recombinant alleles Reject independent assortment Do not accept crossing over of sister chromatids	(1)

Question Number	Answer	Additional guidance	Mark
8(a)(ii)	A calculation showing the following steps: • value for q (1) • value for p² (1) • number of people without a mutated allele for ARD1 (1)	<u>Example of calculation:</u> $\sqrt{(1 \div 300\,000)}$ OR $\sqrt{(26000 \div 7800\,000)}$ = $1.825741858 \times 10^{-3}$ $\{1 - 1.825741858 \times 10^{-3}\}^2 = 0.9981742581$ answer in range 7 768 800 000 to 7 771 545 079 Correct answer on answer line scores full marks	(3)

Question Number	Answer	Additional guidance	Mark
8(b)	An explanation that includes the following points: • (active) mRNA only contains exons whereas pre-mRNA contains both introns and exons / active mRNA doesn't have introns whereas pre-mRNA does (1) • active mRNA may have {fewer / different order of} exons (than pre-mRNA) (1)	accept (active) mRNA is shorter / converse accept pre-mRNA contains a different sequence of bases (in final exon order to active mRNA)	(2)

Question Number	Answer	Additional guidance	Mark
8(c)(i)	An answer that makes reference to the following points: • {pluripotent stem cells / hiPSC} can give {differentiate / specialise / change into} into {most types of / some types of / liver} cells (1) • through differential gene expression (1) • gene for arginase is {active / switched on / not switched off} • by {epigenetic modification / histone modification / (DNA) methylation} (1) • therefore {transcription / translation} occurs (to produce protein) (1)	ignore pluripotent cell becomes a cell that can synthesise arginase do not accept can differentiate into {any / all} type of cell Accept some genes are {switched off / not active / silenced} accept arginase gene expressed accept acetylation accept description e.g. adding methyl groups (to DNA) Accept protein synthesis can occur	(5)

Question Number	Answer	Additional guidance	Mark
8(c)(ii)	An answer that makes reference to three of the following points: • (hiPSC) are {from skin cells / from (another) adult} (1) • (therefore) no {destruction / harming} of {embryos / life} (1) • treatment occurs in {liver/somatic} cells, therefore no changes to {gametes / germ cells} (1) • treatment {is long lasting / doesn't need to be repeated} (1)	accept easier to obtain hiPSC ignore from own body accept embryonic stem cells are not used accept more ethical than using embryonic stem cells / {fewer / no} ethical issues (for hiPSC) Accept there are ethical issues with using embryonic cells accept consent given / converse for embryos ignore no risk of rejection accept to provide treatment for a fatal disease / may be only effective treatment / can save life (of someone with AD) / to reduce death from AD ignore to treat patients with AD	(3)